

hovid-Bisoprolol Fumarate Tablet 10 mg

VIBIS01-0

DESCRIPTION

Round, 8 mm in diameter, light orange film-coated tablet with shallow convex faces, "H" and "D" embossed on each side separated with a score line on the same face.

COMPOSITION

Each tablet contains:
Bisoprolol fumarate 10 mg.

ACTION AND PHARMACOLOGY

Bisoprolol is a highly beta₁-selective-adrenoceptor blocking agent, lacking intrinsic stimulating and relevant membrane stabilising activity. It only shows low affinity to the beta₂-receptor of the smooth muscles of bronchi and vessels as well as to the beta₂-receptors concerned with metabolic regulation. Therefore, bisoprolol is generally not to be expected to influence the airway resistance and beta₂-mediated metabolic effects. Its beta₁-selectivity extends beyond the therapeutic dose range.

PHARMACOKINETICS

Absorption: Bisoprolol is absorbed and has a biological availability of about 90% after oral administration. Peak plasma concentrations occur 2 to 4 hours after oral doses.

Distribution: The distribution volume is 3.5 L/kg. The plasma protein binding of bisoprolol is about 30%. Bisoprolol is moderately lipid-soluble.

Biotransformation and Elimination: Bisoprolol is excreted from the body by two routes. 50% is metabolised by the liver to inactive metabolites which are then excreted by the kidneys. The remaining 50% is excreted by the kidneys in an unmetabolised form. Total clearance is approximately 15 L/h. The half-life in plasma of 10-12 hours gives a 24 hour effect after dosing once daily.

Linearity: The kinetics of bisoprolol are linear and independent of age.

INDICATION

Treatment of high blood pressure (hypertension).

Treatment of coronary heart disease (angina pectoris).

Treatment of stable chronic heart failure with reduced systolic left ventricular function in addition to ACE inhibitors, and diuretics, and optionally cardiac glycosides.

CONTRAINDICATIONS

Bisoprolol is contraindicated in chronic heart failure patients with:

- acute heart failure or during episodes of heart failure decompensation requiring i.v. inotropic therapy
- shock induced by disorders of cardiac function (cardiogenic shock)
- severe disturbances of atrioventricular conduction (second or third degree AV block)
- sick sinus syndrome
- sinoatrial block
- symptomatic bradycardia
- symptomatic hypotension
- severe bronchial asthma
- severe forms of peripheral arterial occlusive disease or severe forms of Raynaud's syndrome
- untreated tumours of the adrenal gland (phaeochromocytoma)
- metabolic acidosis
- hypersensitivity to bisoprolol or to any of the excipients

WARNINGS AND PRECAUTIONS

The treatment of stable chronic heart failure with bisoprolol has to be initiated with a special titration phase.

Especially in patients with ischaemic heart disease the cessation of therapy with bisoprolol must not be done abruptly unless clearly indicated, because this may lead to transitional worsening of heart condition.

The initiation and cessation of treatment with bisoprolol necessitates regular monitoring.

Bisoprolol must be used with caution in:

- bronchospasm (bronchial asthma, obstructive airways diseases)
- diabetes mellitus with large fluctuations in blood glucose values; Symptoms of hypoglycaemia can be masked
- strict fasting
- ongoing desensitisation therapy. As with other beta-blockers, bisoprolol may increase both the sensitivity towards allergens and the severity of anaphylactic reactions. Epinephrine treatment does not always yield the expected therapeutic effect.
- first degree AV block
- Prinzmetal's angina
- peripheral arterial occlusive disease. Aggravation of symptoms may occur especially when starting therapy.
- general anaesthesia

General anaesthesia: In patients undergoing general anaesthesia beta-blockade reduces the incidence of arrhythmias and myocardial ischemia during induction and intubation, and the post-operative period. It is currently recommended that maintenance beta-blockade be continued peri-operatively. The anaesthetist must be aware of beta-blockade because of the potential for interactions with other drugs, resulting in bradyarrhythmias, attenuation of the reflex tachycardia and the decreased reflex ability to compensate for blood loss. If it is thought necessary to withdraw beta-blocker therapy before surgery, this should be done gradually and completed about 48 hours before anaesthesia.

Calcium antagonists: Combination of bisoprolol with calcium antagonists of the verapamil or diltiazem type, with Class I antiarrhythmic drugs and with centrally acting antihypertensive drugs is generally not recommended.

Respiratory system: Although cardioselective (beta₁) beta-blockers may have less effect on lung function than non-selective beta-blockers, as with all beta-blockers, these should be avoided in patients with obstructive airways diseases, unless there are compelling clinical reasons for their use. Where such reasons exist, bisoprolol may be used with caution. In patients with obstructive airways diseases, the treatment with bisoprolol should be started at the lowest possible dose and patients should be carefully monitored for new symptoms (e.g. dyspnea, exercise intolerance, cough). In bronchial asthma or other chronic obstructive lung diseases, which may cause symptoms, bronchodilating therapy should be given concomitantly. Occasionally an increase of the airway resistance may occur in patients with asthma, therefore the dose of beta₂-stimulants may have to be increased.

Psoriasis: Patients with psoriasis or with a history of psoriasis should only be given beta-blockers (e.g. bisoprolol) after carefully balancing the benefits against the risks.

Phaeochromocytoma: In patients with phaeochromocytoma bisoprolol must not be administered until after alpha-receptor blockade.

Thyrotoxicosis: Under treatment with bisoprolol the symptoms of a thyrotoxicosis may be masked.

Special populations: There is no therapeutic experience of bisoprolol treatment of heart failure in patients with the following diseases and conditions.

- insulin dependent diabetes mellitus (type I)
- severely impaired renal function
- severely impaired hepatic function
- restrictive cardiomyopathy
- congenital heart disease
- haemodynamically significant organic valvular disease
- myocardial infarction within 3 months

Effects on ability to drive and use machines: Due to individual variations in reactions to the drug, the ability to drive a vehicle or to operate machinery may be impaired. This should be considered particularly at start of treatment and upon change of medication as well as in conjunction with alcohol.

PREGNANCY AND LACTATION

Pregnancy: Bisoprolol should be given only if the potential benefit justifies the potential risk to the fetus.

Bisoprolol has pharmacological effects that may cause harmful effects on pregnancy and/or the fetus/newborn. In general, beta-adrenoceptor blockers reduce placental perfusion, which has been associated with growth retardation, intrauterine death, abortion or early labour. Adverse effects (e.g. hypoglycaemia and bradycardia) may occur in the fetus and newborn infant. If treatment with beta-adrenoceptor blockers is necessary, beta₁-selective adrenoceptor blockers are preferable.

Bisoprolol should not be used during pregnancy unless clearly necessary. If treatment with bisoprolol is considered necessary, the uteroplacental blood flow and the fetal growth should be monitored. In case of harmful effects on pregnancy or the fetus alternative treatment should be considered. The newborn infant must be closely monitored. Symptoms of hypoglycaemia and bradycardia are generally to be expected within the first 3 days.

Lactation: It is not known whether this drug is excreted in human milk. Therefore, breastfeeding is not recommended during administration of bisoprolol.

DRUG INTERACTIONS**Combinations not recommended:**

- **Calcium antagonists of the verapamil type and to a lesser extent of the diltiazem type:** Negative influence on contractility and atrio-ventricular conduction. Intravenous administration of verapamil in patients on beta-blocker treatment may lead to profound hypotension and atrioventricular block.
- **Class I antiarrhythmic drugs (e.g. quinidine, disopyramide; lidocaine, phenytoin; flecainide, propafenone):** Effect on atrio-ventricular conduction time may be potentiated and negative inotropic effect increased.
- **Centrally acting antihypertensive drugs such as clonidine and others (e.g. methyldopa, moxonidine, rilmenidine):** Concomitant use of centrally acting antihypertensive drugs may worsen heart failure by a decrease in the central sympathetic tone (reduction of heart rate and cardiac output, vasodilation). Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of "rebound hypertension".

Combinations to be used with caution:

- **Calcium antagonists of the dihydropyridine type such as felodipine and amlodipine:** Concomitant use may increase the risk of hypotension, and an increase in the risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.
- **Class-III antiarrhythmic drugs (e.g. amiodarone):** Effect on atrio-ventricular conduction time may be potentiated. Topical beta-blockers (e.g. eye drops for glaucoma treatment) may add to the systemic effects of bisoprolol.
- **Parasympathomimetic drugs:** Concomitant use may increase atrio-ventricular conduction time and the risk of bradycardia.
- **Insulin and oral antidiabetic drugs:** Increase of blood sugar lowering effect. Blockade of beta-adrenoreceptors may mask symptoms of hypoglycaemia.

- **Anaesthetic agents:** Attenuation of the reflex tachycardia and increase of the risk of hypotension.
- **Digitalis glycosides:** Reduction of heart rate, increase of atrio-ventricular conduction time.
- **Non-steroidal anti-inflammatory drugs (NSAIDs):** NSAIDs may reduce the hypotensive effect of bisoprolol.
- **β -Sympathomimetic agents (e.g. Isoprenaline, dobutamine):** Combination with bisoprolol may reduce the effect of both agents.
- **Sympathomimetics that activate both β - and α -adrenoceptors (e.g. noradrenaline, adrenaline):** Combination with bisoprolol may unmask the α -adrenoceptor-mediated vasoconstrictor effects of these agents leading to blood pressure increase and exacerbated intermittent claudication. Such interactions are considered to be more likely with nonselective β -blockers.
- **Concomitant use with antihypertensive agents as well as with other drugs with blood pressure lowering potential (e.g. tricyclic antidepressants, barbiturates, phenothiazines) may increase the risk of hypotension.**

Combinations to be considered

- **Mefloquine:** Increased risk of bradycardia
- **Monoamine oxidase inhibitors (except MAO-B inhibitors):** Enhanced hypotensive effect of the β -blockers but also risk for hypertensive crisis.

MAIN SIDE/ADVERSE EFFECTS

The following definitions apply to the frequency terminology used hereafter:
 Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), Very rare ($< 1/10,000$). Frequency not known (cannot be estimated from available data)

- **Cardiac disorders**
Very common: bradycardia.
Common: worsening of heart failure.
Uncommon: AV-conduction disturbances.
- **Investigations**
Rare: increased triglycerides, increased liver enzymes (ALAT, ASAT).
- **Nervous system disorders**
Common: dizziness, headache.
Rare: syncope
- **Eye disorders**
Rare: reduced tear flow (to be considered if the patient uses lenses).
Very rare: conjunctivitis.
- **Ear and labyrinth disorders**
Rare: hearing disorders.
- **Respiratory, thoracic and mediastinal disorders**
Uncommon: bronchospasm in patients with bronchial asthma or a history of obstructive airways disease.
Rare: allergic rhinitis.
- **Gastrointestinal disorders**
Common: gastrointestinal complaints such as nausea, vomiting, diarrhoea, constipation.
- **Skin and subcutaneous tissue disorders**
Rare: hypersensitivity reactions (itching, flush, rash).
Very rare: alopecia. β -blockers may provoke or worsen psoriasis or induce psoriasis-like rash
- **Musculoskeletal and connective tissue disorders**
Uncommon: muscular weakness and cramps.
- **Vascular disorders**
Common: feeling of coldness or numbness in the extremities, hypotension.
Uncommon: orthostatic hypotension.
- **General disorders**
Common: asthenia, fatigue.
- **Hepatobiliary disorders**
Rare: hepatitis.
- **Reproductive system and breast disorders**
Rare: potency disorders.
- **Psychiatric disorders**
Uncommon: sleep disorders, depression.
Rare: nightmares, hallucinations.

OVERDOSE AND TREATMENT

Symptoms
 In general the most common signs expected with overdosage of a β -blocker are bradycardia, hypotension, bronchospasm, acute cardiac insufficiency and hypoglycaemia.

Treatment
 If overdose occurs, bisoprolol treatment should be stopped and supportive and symptomatic treatment should be provided. Based on the expected pharmacologic actions and recommendations for other β -blockers, the following general measures should be considered when clinically warranted.

Bradycardia: Administer intravenous atropine. If the response is inadequate, isoprenaline or another agent with positive chronotropic properties may be given cautiously. Under some circumstances, transvenous pacemaker insertion may be necessary.

Hypotension: Intravenous fluids and vasopressors should be administered. Intravenous glucagon may be useful.

AV block (second or third degree): Patients should be carefully monitored and treated with isoprenaline infusion or transvenous cardiac pacemaker insertion.

Acute worsening of heart failure: Administer i.v. diuretics, inotropic agents, vasodilating agents.

Bronchospasm: Administer bronchodilator therapy such as isoprenaline, beta2-sympathomimetic drugs and/or aminophylline.

Hypoglycaemia: Administer i.v. glucose.

DOSAGE AND ADMINISTRATION

Method of Administration: For oral use.

Bisoprolol tablets should be taken in the morning and can be taken with food. They should be swallowed with liquid and should not be chewed.

Treatment of hypertension or angina pectoris: In all cases the dose regimen is adjusted individually by your doctor, in particular according to the pulse rate and therapeutic success. The usual initial dose is 5 mg bisoprolol fumarate once daily. If necessary, the dose may be increased to 10 mg bisoprolol fumarate once daily. The maximum recommended dose is 20 mg bisoprolol fumarate once daily. Bisoprolol must be used with caution in patients with hypertension or angina pectoris and accompanying heart failure.

Treatment of stable chronic heart failure (CHF): Standard treatment of CHF consists of an ACE inhibitor (or an angiotensin receptor blocker in case of intolerance to ACE inhibitors), a β -blocker, diuretics, and when appropriate cardiac glycosides. Patients should be stable (without acute failure) when bisoprolol treatment is initiated.

It is recommended that the treating physician should be experienced in the management of chronic heart failure.

Transient worsening of heart failure, hypotension, or bradycardia may occur during the titration period and thereafter.

Titration phase:

The treatment of stable chronic heart failure with bisoprolol requires a titration phase. The treatment with bisoprolol is to be started with a gradual up-titration according to the following steps:

- 1.25 mg once daily for 1 week, if well tolerated increase to *
- 2.5 mg once daily for a further week, if well tolerated increase to *
- 3.75 mg once daily for a further week, if well tolerated increase to *
- 5 mg once daily for the 4 following weeks, if well tolerated increase to
- 7.5 mg once daily for the 4 following weeks, if well tolerated increase to
- 10 mg once daily for the maintenance therapy.

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The maximum recommended dose is 10 mg once daily.

* Bisoprolol 10mg is not suitable for initial treatment of stable chronic heart failure. Lower strengths are available for this purpose.

Close monitoring of vital signs (heart rate, blood pressure) and symptoms of worsening heart failure is recommended during the titration phase. Symptoms may already occur within the first day after initiating the therapy.

Treatment modification:

If the maximum recommended dose is not well tolerated, gradual dose reduction may be considered.

In case of transient worsening of heart failure, hypotension, or bradycardia reconsideration of the dosage of the concomitant medication is recommended. It may also be necessary to temporarily lower the dose of bisoprolol or to consider discontinuation.

The reintroduction and/or up-titration of bisoprolol should always be considered when the patient becomes stable again.

If discontinuation is considered, gradual dose decrease is recommended, since abrupt withdrawal may lead to acute deterioration of the patient's condition.

Duration of treatment:

Treatment of stable chronic heart failure with bisoprolol is generally a long-term treatment.

Additional information:

Do not stop treatment abruptly or change the recommended dose without talking to your doctor first since this might lead to a transitory worsening of heart condition. Especially in patients with ischaemic heart disease, treatment must not be discontinued suddenly. If discontinuation is necessary, the daily dose is gradually decreased.

Patients with hepatic or renal impairment: Treatment of hypertension or angina pectoris: In patients with liver or kidney function disorders of mild to moderate severity no dosage adjustment is normally required. In patients with severe renal impairment (creatinine clearance < 20 ml/min) and in patients with severe hepatic impairment a daily dose of 10 mg bisoprolol fumarate must not be exceeded.

Treatment of stable chronic heart failure: There is no information regarding pharmacokinetics of bisoprolol in patients with chronic heart failure and with impaired hepatic or renal function. Up-titration of the dose in these populations should therefore be made with additional caution.

Elderly people: No dosage adjustment is required.

Paediatric population: There is no paediatric experience with bisoprolol, therefore its use cannot be recommended in paediatric patients.

Note: The information provided here is limited. For further information, kindly consult your doctor or pharmacist.

Storage: Store below 30°C.

Presentation/ Packing:
 Blister pack of 10's, 30's and 100's

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